

surveysd 2.0.0

Overview with a Focus on New Features

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R Package surveysd

- Used in **production** for calibration and survey error estimation at **Statistics Austria**.
- Since 2019 on **CRAN**
- Development version on GitHub:
<https://github.com/statistikat/surveysd>



- R package for the production of survey estimates and survey error estimates
 - Error estimation via calibrated bootstrap replicate weights (Preston, J. (2009))
 - Can deal with panel surveys → **realistic bootstrap draws**
 - Simple but powerful API to construct many estimates with little code
 - Survey calibration with `ipf()`

Workflow

```
1 library(surveysd)
2 eusilc <- demo.eusilc(n = 4, prettyNames = TRUE)
3
4 # 1. draw bootstrap replicates
5 dat_boot <- draw.bootstrap(eusilc, REP = 10, hid = "hid", weights = "pWeight",
6                           strata = "region", period = "year")
7
8 # 2. recalibrate by gender(persons) and region(households)
9 dat_boot_calib <- recalib(dat_boot, conP.var = "gender", conH.var = "region")
10
11 # 3. estimate weightedRatio for povertyRisk per period and group
12 err.est <- calc.stError(dat_boot_calib,
13                         var = "povertyRisk",
14                         group = "gender",
15                         fun = weightedRatio)
16 err.est$Estimates # large table with all estimates
```


A photograph of a modern building's interior, featuring multiple levels with glass railings and potted plants. The image is overlaid with a semi-transparent blue filter. On the right side, there is a vertical strip showing a closer view of the building's glass facade and windows.

Improvements with surveysd 2.0.0

Additional bootstrap replicate method

- Implemented **Rao-Wu** replicates (Rao, J. N. K., and C. F. J. Wu. (1988)) and included **vignette**

```
1 dat_boot_rw <- draw.bootstrap(eusilc,  
2                               method = "Rao-Wu",  
3                               REP = 10,  
4                               hid = "hid",  
5                               weights = "pWeight",  
6                               strata = "region",  
7                               period = "year")
```

- Applicable if sampling fraction is small (<10%)

Improvements with drawing bootstraps & calibration and estimation step

- Improved *panel* behaviour within bootstrap replicates → sampling proportions are more stable over panel waves.
- Improvements with calibration
 - underlying calibration: `ipf()` - iterative proportional fitting
 - Calibration for numerical variables (e.g. enterprise turnover, personal income, ...) more stable

Improvements with estimation step

- Difference of estimates between groups → `group.diff`
 - Compatible with panel surveys, k-mean averages, ...

```
1 dat_subset <- dat_boot_calib[year == 2010]
2
3 povertyRates <- calc.stError(dat_subset, var = "povertyRisk",
4                               fun = weightedRatio,
5                               group = c("gender", "region"),
6                               group.diff = TRUE)
7
8 # filter results for group differences
9 povertyRates$Estimates[estimate_type == "group difference"]
```


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